

NVIDIA FY2025 Q2 earnings call transcript

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Nvidia (NASDAQ: NVDA)

Q2 2025 Earnings Call

Aug 28, 2024 , 5:00 p.m. ET

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Operator

Good afternoon. My name is Abby, and I will be your conference operator today. At this time, I would like to welcome everyone to NVIDIA's second-quarter earnings call. All lines have been placed on mute to prevent any background noise.

After the speakers' remarks, there will be a question-and-answer session. [Operator instructions] Thank you. And Mr. Stewart Stecker, you may begin your conference.

Stewart Stecker -- Senior Director, Investor Relations

Thank you. Good afternoon, everyone, and welcome to NVIDIA's conference call for the second quarter of fiscal 2025. With me today from NVIDIA are Jensen Huang, president and chief executive officer; and Colette Kress, executive vice president and chief financial officer. I would like to remind you that our call is being webcast live on NVIDIA's Investor Relations website.

The webcast will be available for replay until the conference call to discuss our financial results for the third quarter of fiscal 2025. The content of today's call is NVIDIA's property. It cannot be reproduced or transcribed without prior written consent. During this call, we may make forward-looking statements based on current expectation.

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These are subject to a number of risks, significant risks, and uncertainties, and our actual results may differ materially. For a discussion of factors that could affect our future financial results and business, please refer to the disclosure in today's earnings release, our most recent Forms 10-K and 10-Q, and the reports that we may file on Form 8-K with the Securities and Exchange Commission. All our statements are made as of today, August 28, 2024, based on information currently available to us. Except as required by law, we assume no obligation to update any such statements.

During this call, we will discuss non-GAAP financial measures. You can find a reconciliation of these non-GAAP financial measures to GAAP financial measures in our CFO commentary, which is posted on our website. Let me highlight an upcoming event for the financial community. We will be attending the Goldman Sachs Communacopia and Technology Conference on September 11 in San Francisco, where Jensen will participate in a keynote fireside chat.

Our earnings call to discuss the results of our third quarter of fiscal 2025 is scheduled for Wednesday, November 20, 2024. With that, let me turn the call over to Colette.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Thanks, Stewart. Q2 was another record quarter. Revenue of \$30 billion was up 15% sequentially and up 122% year on year and well above our outlook of \$28 billion. Starting with Data Center.

Data Center revenue of \$26.3 billion was a record, up 16% sequentially and up 154% year on year, driven by strong demand for NVIDIA Hopper, GPU computing, and our networking platforms. Compute revenue grew more than 2.5x. Networking revenue grew more than 2x from the last year. Cloud service providers represented roughly 45% of our Data Center revenue, and more than 50% stemmed from the consumer Internet and enterprise companies.

Customers continue to accelerate their Hopper architecture purchases while gearing up to adopt Blackwell. Key workloads driving our Data Center growth include generative AI model training and inferencing; video, image, and text data pre and post processing with CUDA and AI workloads; synthetic data generation; AI-powered recommender systems; SQL and Vector database processing as well. Next-generation models will require 10 to 20 times more compute to train with significantly more data. The trend is expected to continue.

Over the trailing four quarters, we estimate that inference drove more than 40% of our Data Center revenue. CSPs, consumer Internet companies, and enterprises benefit from the incredible throughput and efficiency of NVIDIA's inference platform. Demand for NVIDIA is coming from frontier model makers, consumer Internet services, and tens of thousands of companies and start-ups building generative AI applications for consumers, advertising, education, enterprise and healthcare, and robotics. Developers desire NVIDIA's rich ecosystem and availability in every cloud.

CSPs appreciate the broad adoption of NVIDIA and are growing their NVIDIA capacity given the high demand. NVIDIA H200 platform began ramping in Q2, shipping to large CSPs, consumer Internet, and enterprise company. The NVIDIA H200 builds upon the strength of our Hopper architecture and offering over 40% more memory bandwidth compared to the H100. Our Data Center revenue in China grew sequentially in Q2 and a significant contributor to our Data Center revenue.

As a percentage of total Data Center revenue, it remains below levels seen prior to the imposition of export controls. We continue to expect the China market to be very competitive going forward. The latest round of MLPerf inference benchmarks highlighted NVIDIA's inference

leadership with both NVIDIA Hopper and Blackwell platform combining to win gold medals on all tasks. At Computex, NVIDIA, with the top computer manufacturers, unveiled an array of Blackwell architecture-powered systems and NVIDIA networking for building AI factories and data centers.

With the NVIDIA MGX modular reference architecture, our OEMs and ODM partners are building more than 100 Blackwell-based systems designed quickly and cost-effectively. The NVIDIA Blackwell platform brings together multiple GPU, CPU, DPU, NVLink, and Link Switch and the networking chips, systems, and NVIDIA CUDA software to power the next generation of AI across the cases, industries, and countries. The NVIDIA GB200 NVL72 system with the fifth-generation NVLink enables all 72 GPUs to act as a single GPU and deliver up to 30x faster inference for LLM's workloads and unlocking the ability to run trillion-parameter models in real time. Hopper demand is strong, and Blackwell is widely sampling.

We executed a change to the Blackwell GPU mass to improve production yields. Blackwell production ramp is scheduled to begin in the fourth quarter and continue into fiscal year '26.

In Q4, we expect to get several billion dollars in Blackwell revenue. Hopper shipments are expected to increase in the second half of fiscal 2025.

Hopper supply and availability have improved. Demand for Blackwell platforms is well above supply, and we expect this to continue into next year. Networking revenue increased 16% sequentially. Our Ethernet for AI revenue, which includes our Spectrum-X end-to-end Ethernet platform, doubled sequentially with hundreds of customers adopting our Ethernet offerings.

Spectrum-X has broad market support from OEM and ODM partners and is being adopted by CSPs, GPU cloud providers, and enterprises, including xAI to connect the largest GPU compute cluster in the world. Spectrum-X supercharges Ethernet for AI processing and delivers 1.6x the performance of traditional Ethernet. We plan to launch new Spectrum-X products every year to support demand for scaling compute clusters from tens of thousands of GPUs today to millions of DPUs in the near future. Spectrum-X is well on track to begin a multibillion-dollar product line within a year.

Our sovereign AI opportunities continue to expand as countries recognize AI expertise and infrastructure at national imperatives for their society and industries. Japan's National Institute of Advanced Industrial Science and Technology is building its AI Bridging Cloud Infrastructure 3.0 supercomputer with NVIDIA. We believe sovereign AI revenue will reach low double-digit billions this year. The enterprise AI wave has started.

Enterprises also drove sequential revenue growth in the quarter. We are working with most of the Fortune 100 companies on AI initiatives across industries and geographies. A range of applications are fueling our growth, including AI-powered chatbots, generative AI copilots, and agents to build new monetizable business applications and enhance employee productivity. Amdocs is using NVIDIA generative AI for their smart agent, transforming the customer experience and reducing customer service costs by 30%.

ServiceNow is using NVIDIA for its Now Assist offering, the fastest-growing new product in the company's history. SAP is using NVIDIA to build dual copilots. Cohesity is using NVIDIA to build their generative AI agent and lower generative AI development costs. Snowflake, serves over 3 billion queries a day for over 10,000 enterprise customers, is working with NVIDIA to build copilots.

And lastly, is using NVIDIA AI Omniverse to reduce end-to-end cycle times for their factories by 50%. Automotive was a key growth driver for the quarter as every automaker developing autonomous vehicle technology is using NVIDIA in their data centers. Automotive will drive multibillion dollars in revenue across on-prem and cloud consumption and will grow as next-generation AV models require significantly more compute. Health care is also on its way to being a multibillion-dollar business as AI revolutionizes medical imaging, surgical robots, patient care, electronic health record processing, and drug discovery.

During the quarter, we announced a new NVIDIA AI foundry service to supercharge generative AI for the world's enterprises with Meta's Llama 3.1 collection of models. This marks a watershed moment for enterprise AI. Companies for the first time can leverage the capabilities of an open-source frontier-level model to develop customized AI applications to encode their institutional knowledge into an AI flywheel to automate and accelerate their business. Accenture is the first to adopt the new service to build custom Llama 3.1 models for both its own use and to assist clients seeking to deploy generative AI applications.

NVIDIA NIMs accelerate and simplify model deployment. Companies across healthcare, energy, financial services, retail, transportation, and telecommunications are adopting NIMs, including Aramco, Lowe's, and Uber. AT&T realized 70% cost savings and eight times latency reduction after moving into NIMs for generative AI, call transcription, and classification. Over 150 partners are embedding NIMs across every layer of the AI ecosystem.

We announced NIM Agent Blueprint, a catalog of customizable reference applications that include a full suite of software for building and deploying enterprise generative AI applications. With NIM Agent Blueprint, enterprises can refine their AI applications over time, creating a data-driven AI flywheel. The first NIM Agent Blueprints include workloads for customer service, computer-aided drug discovery, and enterprise retrieval augmented generation. Our system integrators, technology solution providers, and system builders are bringing NVIDIA NIM Agent Blueprints to enterprises.

NVIDIA NIM and NIM Agent Blueprints are available through the NVIDIA AI Enterprise software platform, which has great momentum. We expect our software, SaaS, and support revenue to approach a \$2 billion annual run rate exiting this year, with NVIDIA AI Enterprise notably contributing to growth. Moving to gaming and AI PC. Gaming revenue of \$2.88 billion increased 9% sequentially and 16% year on year.

We saw sequential growth in console, notebook, and desktop revenue, and demand is strong and growing and channel inventory remains healthy. Every PC with RTX is an AI PC. RTX PCs can deliver up to 1,300 AI tops and are now over 200 RTX AI laptops designed from leading PC manufacturers. With 600 AI-powered applications and games and an installed base of 100 million devices, RTX is set to revolutionize consumer experiences with generative AI.

NVIDIA ACE, a suite of generative AI technologies is available for RTX AI PCs. Megabreak is the first game to use NVIDIA ACE, including our small language model, Nemotron 4B optimized on device inference. The NVIDIA gaming ecosystem continues to grow. Recently added RTX and DLSS titles include Indiana Jones and The Great Circle, Awakening, and Dragon Age: The Vanguard.

The GeForce NOW library continues to expand with total catalog size of over 2,000 titles, the most content of any cloud gaming service. Moving to pro visualization. Revenue of \$454 million was up 6% sequentially and 20% year on year. Demand is being driven by AI and graphic use cases, including model fine-tuning and Omniverse-related workloads.

Automotive and manufacturing were among the key industry verticals driving growth this quarter. Companies are racing to digitalize workflows to drive efficiency across their operations. The world's largest electronics manufacturer, Foxconn, is using NVIDIA Omniverse to power digital twins of the physical plants that produce NVIDIA Blackwell systems. And several large global enterprises, including Mercedes-Benz, signed multiyear contracts for NVIDIA Omniverse Cloud to build industrial digital twins of factories.

We announced new NVIDIA USD NIMs and connectors to open Omniverse to new industries and enable developers to incorporate generative AI copilots and agents into USD workloads, accelerating our ability to build highly accurate virtual worlds. WPP is implementing the USD NIM microservices in its generative AI-enabled content creation pipeline for customers such as The Coca-Cola Company. Moving to automotive and robotics. Revenue was \$346 million, up 5% sequentially and up 37% year on year.

Year-on-year growth was driven by the new customer ramp in self-driving platforms and increased demand for AI cockpit solutions. At the consumer -- at the Computer Vision and Pattern Recognition Conference, NVIDIA won the Autonomous Brand Challenge in the end-to-end driving upscale category, outperforming more than 400 entries worldwide. Boston Dynamics, BYD Electronics, Figure, Intrinsic, Siemens, and Teradyne Robotics are using the NVIDIA Isaac robotics platform for autonomous robot arms, humanoids, and mobile robots. Now, moving to the rest of the P&L.

GAAP gross margins were 75.1% and non-GAAP gross margins were 75.7%, down sequentially due to a higher mix of new products within Data Center and inventory provisions for low-yielding Blackwell material. Sequentially, GAAP and non-GAAP operating expenses were up 12%, primarily reflecting higher compensation-related costs. Cash flow from operations was \$14.5 billion. In Q2, we utilized cash of \$7.4 billion toward shareholder returns in the form of share repurchases and cash dividends, reflecting the increase in dividend per shareholder.

Our board of directors recently approved a \$50 billion share repurchase authorization to add to our remaining \$7.5 billion of authorization at the end of Q2. Let me turn the outlook for the third quarter. Total revenue is expected to be \$32.5 billion, plus or minus 2%. Our third-quarter revenue outlook incorporates continued growth of our Hopper architecture and sampling of our Blackwell products.

We expect Blackwell production ramp in Q4. GAAP and non-GAAP gross margins are expected to be 74.4% and 75%, respectively, plus or minus 50 basis points. As our Data Center mix continues to shift to new products, we expect this trend to continue into the fourth quarter of fiscal 2025. For the full year, we expect gross margins to be in the mid-70% range.

GAAP and non-GAAP operating expenses are expected to be approximately \$4.3 billion and \$3.0 billion, respectively. Full-year operating expenses are expected to grow in the mid- to upper 40% range as we work on developing our next generation of products. GAAP and non-GAAP other income and expenses are expected to be about \$350 million, including gains and losses from nonaffiliated investments and publicly held equity securities. GAAP and non-GAAP tax rates are expected to be 17%, plus or minus 1%, excluding any discrete items.

Further financial details are included in the CFO commentary and other information available on our IR website. We are now going to open the call for questions. Operator, would you please help us poll for questions?

Questions & Answers:

Operator

Thank you. [Operator instructions] We will pause for just a moment to compile the Q&A roster. And as a reminder, we ask that you please limit yourself to one question. And your first question comes from the line of Vivek Arya with Bank of America Securities.

Your line is open.

Vivek Arya -- Analyst

Thanks for taking my question. Jensen, you mentioned in the prepared comments that there's a change in the Blackwell GPU mask. I'm curious, are there any other incremental changes in back-end packaging or anything else? And I think related, you suggested that you could ship several billion dollars of Blackwell in Q4 despite the change in the design. Is it because all these issues will be solved by then? Just help us size what is the overall impact of any changes in Blackwell timing, what that means to your kind of revenue profile and how are customers reacting to it.

Jensen Huang -- President and Chief Executive Officer

Yeah. Thanks, Vivek. The change to the mask is complete. There were no functional changes necessary.

And so, we're sampling functional samples of Blackwell, Grace Blackwell, and a variety of system configurations as we speak. There are something like 100 different types of Blackwell-based systems that are built that were shown at Computex, and we're enabling our ecosystem to start sampling those. The functionality of Blackwell is as it is, and we expect to start production in Q4.

Operator

And your next question comes from the line of Toshiya Hari with Goldman Sachs. Your line is open.

Toshiya Hari -- Analyst

Hi. Thank you so much for taking the question. Jensen, I had a relatively longer-term question. As you may know, there's a pretty heated debate in the market on your customers and customers' customers return on investment and what that means for the sustainability of capex going forward.

Internally at NVIDIA, like what are you guys watching? What's on your dashboard as you try to gauge customer return and how that impacts capex? And then a quick follow-up maybe for Colette. I think your sovereign AI number for the full year went up maybe a couple of billion. What's driving the improved outlook and how should we think about fiscal '26? Thank you.

Jensen Huang -- President and Chief Executive Officer

Thanks, Toshiya. First of all, when I said ship production in Q4, I mean shipping out, I don't mean starting to ship, but I mean -- I don't mean starting production but shipping up. On the longer-term question, let's take a step back. And you've heard me say that we're going through two simultaneous platform transitions at the same time.

The first one is transitioning from accelerated computing to -- from general-purpose computing to accelerated computing. And the reason for that is because CPU scaling has been known to be slowing for some time and it has slowed to a crawl. And yet the amount of computing demand continues to grow quite significantly. You could maybe even estimate it to be doubling every single year.

And so, if we don't have a new approach, computing inflation would be driving up the cost for every company, and it would be driving up the energy consumption of data centers around the world. In fact, you're seeing that. And so, the answer is accelerated computing. We know that accelerated computing, of course, speeds up applications.

It also enables you to do computing at a much larger scale, for example, scientific simulations or database processing, but what that translates directly to is lower cost and lower energy consumed. And in fact, this week, there's a blog that came out that talked about a whole bunch of new libraries that we offer. And that's really the core of the first platform transition, going from general-purpose computing to accelerated computing. And it's not unusual to see someone save 90% of their computing cost.

And the reason for that is, of course, you just sped up an application 50x. You would expect the computing cost to decline quite significantly. The second was enabled by accelerated computing because we drove down the cost of training large language models or training deep learning so incredibly that it is now possible to have gigantic scale models, multitrillion-parameter models and train it on -- pretrain it on just about the world's knowledge corpus and

let the model go figure out how to understand human language representation and how to codify knowledge into its neural networks and how to learn reasoning, and so which caused the generative AI revolution. Now, generative AI, taking a step back about why it is that we went so deeply into it is because it's not just a feature, it's not just the capability.

It's a fundamental new way of doing software. Instead of human-engineered algorithms, we now have data. We tell the AI, we tell the model, we tell the computer what are the expected answers. What are our previous observations? And then for it to figure out what the algorithm is, what's the function.

It learns a universal -- AI is a bit of a universal function approximator and it learns the function. And so, you could learn the function of almost anything. And anything that you have that's predictable, anything that has structure, anything that you have previous examples of. And so, now here we are with generative AI.

It's a fundamental new form of computer science. It's affecting how every layer of computing is done from CPU to GPU, from human-engineered algorithms to machine-learned algorithms, and the type of applications you could now develop and produce is fundamentally remarkable. And there are several things that are happening in generative AI. So, the first thing that's happening is the frontier models are growing in quite substantial scale.

And they're still seeing -- we're still all seeing the benefits of scaling. And whenever you double the size of a model, you also have to more than double the size of the data set to go train it. And so, the amount of flops necessary in order to create that model goes up quadratically. And so, it's not unexpected to see that the next-generation models could take 10, 20, 40 times more compute than last generation.

So, we have to continue to drive the generational performance up quite significantly so we can drive down the energy consumed and drive down the cost necessary to do it. And so, the first one is there are larger frontier models trained on more modalities. And surprisingly, there are more frontier model makers than last year. And so, you have more and more and more.

That's one of the dynamics going on in generative AI. The second is although it's below the tip of the iceberg, what we see are ChatGPT image generators. We see coding. We use generative AI for coding quite extensively here at NVIDIA now.

We, of course, have a lot of digital designers and things like that. But those are kind of the tip of the iceberg. What's below the iceberg are the largest systems, largest computing systems in the world today, which are -- and you've heard me talk about this in the past, which are recommender systems moving from CPUs. It's now moving from CPUs to generative AI.

So, recommender systems, ad generation, custom ad generation targeting ads at very large scale and quite hyper-targeting, search, and user-generated content, these are all very large-scale applications have now evolved to generative AI. Of course, the number of generative AI start-ups is generating tens of billions of dollars of cloud renting opportunities for our cloud partners. And sovereign AI, countries that are now realizing that their data is their natural and national resource and they have to use AI, build their own AI infrastructure so that they could have their own digital intelligence. Enterprise AI, as Colette mentioned earlier, is starting, and you might have seen our announcement that the world's leading IT companies are joining us to take the NVIDIA AI Enterprise platform to the world's enterprises.

The companies that we're talking to, so many of them are just so incredibly excited to drive more productivity out of the company. And then general robotics. The big transformation last year as we are able to now learn physical AI from watching video and human demonstration and synthetic data generation from reinforcement learning from systems like Omniverse, we are now able to work with just about every robotics companies now to start thinking about, start building general robotics. And so, you can see that there are just so many different directions that generative AI is going.

And so, we're actually seeing the momentum of generative AI accelerating.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

And Toshiya, to answer your question regarding sovereign AI and our goals in terms of growth, in terms of revenue, it certainly is a unique and growing opportunity, something that surfaced with generative AI and the desires of countries around the world to have their own generative AI that would be able to incorporate their own language, incorporate their own culture, incorporate their own data in that country. So, more and more excitement around these models and what they can be specific for those countries. So, yes, we are seeing some growth opportunity in front of us.

Operator

And your next question comes from the line of Joe Moore with Morgan Stanley. Your line is open.

Joe Moore -- Analyst

Great. Thank you. Jensen, in the press release, you talked about Blackwell anticipation being incredible. But it seems like Hopper demand is also really strong.

I mean, you're guiding for a very strong quarter without Blackwell in October. So, how long do you see sort of coexisting strong demand for both? And can you talk about the transition to Blackwell? Do you see people intermixing clusters? Do you think most of the Blackwell activities, new clusters? Just some sense of what that transition looks like.

Jensen Huang -- President and Chief Executive Officer

Yeah. Thanks, Joe. The demand for Hopper is really strong. And it's true, the demand for Blackwell is incredible.

There's a couple of reasons for that. The first reason is if you just look at the world's cloud service providers, the amount of GPU capacity they have available, it's basically none. And the reason for that is because they're either being deployed internally for accelerating their own workloads, data processing, for example. Data processing, we hardly ever talk about it because it's mundane.

It's not very cool because it doesn't generate a picture or generate words. But almost every single company in the world processes data in the background. And NVIDIA's GPUs are the only accelerators on the planet that process and accelerate data. SQL data, Panda's data, data science toolkits like Panda's, and the new one, Polar's.

These are the ones -- the most popular data processing platforms in the world. And aside from CPUs, which as I've mentioned before, really running out of steam, NVIDIA's accelerated computing is really the only way to get boosting performance out of that. And so, number one is the primary -- the No. 1 use case long before generative AI came along is that the migration of applications one after another to accelerated computing.

The second is, of course, the rentals. They're renting capacity to model makers. They're renting it to start-up companies. And a generative AI company spends the vast majority of their invested capital into infrastructure so that they could use an AI to help them create products.

And so, these companies need it now. They just simply can't afford -- you just raise money. They want you to put it to use now. You have processing that you have to do.

You can't do it next year, you got to do it today. And so, there's a fair -- that's one reason. The second reason for Hopper demand right now is because of the race to the next plateau. The first person to the next plateau gets to be -- get to introduce a revolutionary level of AI.

The second person who gets there is incrementally better or about the same. And so, the ability to systematically and consistently race to the next plateau and be the first one there is how you establish leadership. NVIDIA is constantly doing that, and we show that to the world and the GPUs we make and the AI factories that we make, the networking systems that we make, the SoCs we create. I mean, we want to set the pace.

We want to be consistently the world's best. And that's the reason why we drive ourselves so hard. Of course, we also want to see our dreams come true and all of the capabilities that we imagine in the future and the benefits that we can bring to society, we want to see all that come true. And so, these model makers are the same.

Of course, they want to be the world's best. They want to be the world's first. And although Blackwell will start shipping out in billions of dollars at the end of this year, the standing up of the capacity is still probably weeks and a month or so away. And so, between now and then is a lot of generative AI market dynamic.

And so, everybody is just really in a hurry. It's either operational reasons that they need it. They need accelerated computing. They don't want to build any more general-purpose computing infrastructure and even Hopper.

Of course, H200 is state-of-the-art. Hopper, if you have a choice between building CPU infrastructure right now for business or Hopper infrastructure for business right now, that decision is relatively clear. And so, I think people are just clamoring to transition the \$1 trillion of established installed infrastructure to a modern infrastructure and Hopper's state-of-the-art.

Operator

And your next question comes from the line of Matt Ramsay with TD Cowen. Your line is open.

Matt Ramsay -- Analyst

Thank you very much. Good afternoon, everybody. Jensen, I wanted to kind of circle back to an earlier question about the debate that investors are having about the ROI on all of this capex. And hopefully, this question and the distinction will make some sense.

But what I'm having discussions about is with like the percentage of folks that you see that are spending all of this money and looking to sort of push the frontier toward AGI convergence and, as you just said, a new plateau in capability, and they're going to spend regardless to get to that level of capability because it opens up so many doors for the industry and for their company versus customers that are really, really focused today on capex versus ROI. I don't know if that distinction makes sense. I'm just trying to get a sense of how you're seeing the priorities of people that are putting the dollars in the ground on this new technology and what their priorities are and their time frames are for that investment. Thanks.

Jensen Huang -- President and Chief Executive Officer

Thanks, Matt. The people who are investing in NVIDIA infrastructure are getting returns on it right away. It's the best ROI infrastructure, computing infrastructure investment you can make today. And so, one way to think through it, probably the most -- the easiest way to think through it is just go back to first principles.

You have \$1 trillion worth of general-purpose computing infrastructure. And the question is, do you want to build more of that or not? And for every \$1 billion worth of Juniper CPU-based infrastructure that you stand up, you probably rent it for less than \$1 billion. And so, because it's commoditized, there's already \$1 trillion on the ground. What's the point of getting more? And so, the people who are clamoring to get this infrastructure, one, when they build out Hopper-based infrastructure and soon, Blackwell-based infrastructure, they start saving money.

That's a tremendous return on investment. And the reason why they start saving money is because data processing saves money, and data processing is probably just a giant part of it already. And so, recommender systems save money, so on and so forth, OK? And so, you start saving money. The second thing is everything you stand up are going to get rented because so many companies are being founded to create generative AI.

And so, your capacity gets rented right away and the return on investment of that is really good. And then the third reason is your own business. Do you want to either create the next frontier yourself or your own Internet services benefit from a next-generation ad system or a next-generation recommender system or a next-generation search system? So, for your own services, for your own stores, for your own user-generated content, social media platforms, for your own services, generative AI is also a fast ROI. And so, there's a lot of ways you could think through it.

But at the core, it's because it is the best computing infrastructure you could put in the ground today. The world of general-purpose computing is shifting to accelerated computing. The world of human-engineered software is moving to generative AI software. If you were to build infrastructure to modernize your cloud and your data centers, build it with accelerated computing NVIDIA.

That's the best way to do it.

Operator

And your next question comes from the line of Timothy Arcuri with UBS. Your line is open.

Timothy Arcuri -- Analyst

Thanks a lot. I had a question on the shape of the revenue growth, both near and longer term. I know Colette, you did increase opex for the year. And if I look at the increase in your purchase commitments and your supply obligations, that's also quite bullish.

On the other hand, there's some school of thought that not that many customers really seem ready for liquid cooling, and I do recognize that some of these racks can be air-cooled. But Jensen, is that something to consider sort of on the shape of how Blackwell is going to ramp? And then I guess when you look beyond next year, which is obviously going to be a great year and you look into '26, do you worry about any other gating factors like, say, the power supply chain or at some point, models start to get smaller? I'm just wondering if you can speak to that. Thanks.

Jensen Huang -- President and Chief Executive Officer

I'm going to work backwards. I really appreciate the question, Tim. So, remember, the world is moving from general-purpose computing to accelerated computing. And the world builds about \$1 trillion worth of data centers.

\$1 trillion worth of data centers in a few years will be all accelerated computing. In the past, no GPUs are in data centers, just CPUs. In the future, every single data center will have GPUs. And the reason for that is very clear: because we need to accelerate workloads so that we can continue to be sustainable, continue to drive down the cost of computing so that

when we do more computing, we don't experience computing inflation.

Second, we need GPUs for a new computing model called generative AI that we could all acknowledge is going to be quite transformative to the future of computing. And so, I think working backwards, the way to think about that is the next \$1 trillion of the world's infrastructure will clearly be different than the last \$1 trillion, and it will be vastly accelerated. With respect to the shape of our ramp, we offer multiple configurations of Blackwell. Blackwell comes in either a Blackwell classic, if you will, that uses the HGX form factor that we pioneered with Volta.

And I think it was Volta. And so, we've been shipping the HGX form factor for some time. It is air-cooled. The Grace Blackwell is liquid-cooled.

However, the number of data centers that want to go to liquid-cooled is quite significant. And the reason for that is because we can, in a liquid-cooled data center, in any data center -- power-limited data center, whatever size data center you choose, you could install and deploy anywhere from three to five times the AI throughput compared to the past. And so, liquid cooling is cheaper. Liquid cooling, our TCO is better, and liquid cooling allows you to have the benefit of this capability we call NVLink, which allows us to expand it to 72 Grace Blackwell packages, which has essentially 144 GPUs.

And so, imagine 144 GPUs connected in NVLink. And that, we're increasingly showing you the benefits of that. And the next click is obviously very low latency, very high throughput large language model inference, and the large NVLink domain is going to be a game changer for that. And so, I think people are very comfortable deploying both.

And so, almost every CSP we're working with are deploying some of both. And so, I'm pretty confident that we'll ramp it up just fine. Your second question out of the third is that looking forward, yes, next year is going to be a great year. We expect to grow our Data Center business quite significantly next year.

Blackwell is going to be a complete game changer for the industry. And Blackwell is going to carry into the following year. And as I mentioned earlier, working backwards from first principles, remember that computing is going through two platform transitions at the same time. And that's just really, really important to keep your head on -- your mind focused on, which is general-purpose computing is shifting to accelerated computing, and human-engineered software is going to transition to generative AI or artificial intelligence-learned software.

OK.

Operator

And your next question comes from the line of Stacy Rasgon with Bernstein Research. Your line is open.

Stacy Rasgon -- Analyst

Hi, guys. Thanks for taking my question. So, I have two short questions for Colette. The first several billion dollars of Blackwell revenue in Q4, is that additive? You said you expected Hopper demand to strengthen in the second half.

Does that mean Hopper strengthens Q3 to Q4 as well on top of Blackwell adding several billion dollars? And the second question on gross margins. If I have mid-70s for the year, let's say, where I want to draw that, if I have 75 for the year, I'd be something like 71 to 72 for Q4, somewhere in that range. Is that the kind of exit rate for gross margins that you're expecting? And how should we think about the drivers of gross margin evolution into next year as Blackwell ramps? And I mean, hopefully, I guess the yields and the inventory reserves and everything come up.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

So, Stacy, let's first take your question that you had about Hopper and Blackwell. So, we believe our Hopper will continue to grow into the second half. We have many new products for Hopper, our existing products for Hopper that we believe will start continuing to ramp in the next quarters, including our Q3 and those new products moving to Q4. So, let's say, Hopper there for versus H1 is a growth opportunity for that.

Additionally, we have the Blackwell on top of that, and the Blackwell starting ramping in Q4. So, I hope that helps you on those two pieces. Your second piece is in terms of our gross margin. We provided gross margin for our Q3.

We provided our gross margin on a non-GAAP at about 75. We'll work with all the different transitions that we're going through, but we do believe we can do that 75 in Q3. We provided that we're still on track for the full year also in the mid-70s or approximately the 75. So, we're going to see some slight difference possibly in Q4, again with our transitions and the different cost structures that we have on our new product introductions.

However, I'm not in the same number that you are there. We don't have exactly guidance, but I do believe you're lower than where we are.

Operator

And your next question comes from the line of Ben Reitzes with Melius. Your line is open.

Ben Reitzes -- Melius Research -- Analyst

Yeah. Hey, thanks a lot for the question. I wanted to ask about the geographies. There was the 10-Q that came out, and the United States was down sequentially while several Asian geographies were up a lot sequentially.

Just wondering what the dynamics are there. And obviously, China did very well. You mentioned it in your remarks. What are the puts and takes? And then I just wanted to clarify from Stacy's question if that means the sequential overall revenue growth rates for the company accelerate in the fourth quarter, given all those favorable revenue dynamics.

Thanks.

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Let me talk about a bit in terms of our disclosure in terms of the 10-Q, a required disclosure in a choice of geographies. Very challenging sometimes to create that right disclosure as we have to come up with one key piece. The pieces in terms of we have in terms of who we sell to and/or specifically who we invoice to, and so what you're seeing in terms of there is who we invoice. That's not necessarily where the product will eventually be and where it may even travel to the end customer.

These are just moving to our OEMs or ODMs and our system integrators for the most part across our product portfolio. So, what you're seeing there is sometimes just a swift shift in terms of who they are using to complete their full configuration before those things are going into the data center, going into notebooks and those pieces of it. And that shift happens from time to time. But yes, our China number there are invoicing to China.

Keep in mind that is incorporating both gaming, also Data Center, also automotive in those numbers that we have. Going back to your statement and regarding gross margin and also what we're seeing in terms of what we're looking at for Hopper and Blackwell in terms of revenue. Hopper will continue to grow in the second half. We'll continue to grow from what we are

currently seeing.

Determining that exact mix in each Q3 and Q4, we don't have here. We are not here to guide yet in terms of Q4. But we do see right now the demand expectations. We do see the visibility that that will be a growth opportunity in Q4.

On top of that, we will have our Blackwell architecture.

Operator

And your next question comes from the line of C.J. Muse with Cantor Fitzgerald. Your line is open.

C.J. Muse -- Analyst

Yeah. Good afternoon. Thank you for taking the question. You've embarked on a remarkable annual product cadence with challenges only likely becoming more and more, given rising complexity in a rather limit advanced package world.

So, curious, if you take a step back, how does this backdrop alter your thinking around potentially greater vertical integration, supply chain partnerships, and then taking through a consequential impact to your margin profile? Thank you.

Jensen Huang -- President and Chief Executive Officer

Yeah. Thanks. Let's see. I think the first answer to your -- the answer to your first question is that the reason why our velocity is so high is simultaneously because the complexity of the model is growing, and we want to continue to drive its cost down.

It's growing so we want to continue to increase its scale. And we believe that by continuing to scale the AI models, that we'll reach a level of extraordinary usefulness and that it would open up, realize the next industrial revolution. We believe it. And so, we're going to drive ourselves really hard to continue to go up that scale.

We have the ability, fairly uniquely, to integrate, to design an AI factory because we have all the parts. It's not possible to come up with a new AI factory every year unless you have all the parts. And so, we have -- next year, we're going to ship a lot more CPUs than we've ever had in the history of our company, more GPUs, of course, but also NVLink switches, CX DPUs, ConnectX for East and West, BlueField DPUs for North and South, and data and storage processing to InfiniBand for supercomputing centers, to Ethernet, which is a brand-new product for us, which is well on its way to becoming a multibillion-dollar business to bring AI to Ethernet. And so, the fact that we could build -- we have access to all of this, we have one architectural stack, as you know, it allows us to introduce new capabilities to the market as we complete it.

Otherwise, what happens, you ship these parts, you go find customers to sell it to, and then you've got to build -- somebody's got to build up an AI factory, and the AI factory has got a mountain of software. And so, it's not about who integrates it. We love the fact that our supply chain is disintegrated in the sense that we could service Quanta, Foxconn, HP, Dell, Lenovo, Super Micro. We used to be able to serve ZTE.

They were recently purchased and so on and so forth. And so, the number of ecosystem partners that we have, Gigabyte, the number of ecosystem partners that we have that allows them to take our architecture, which all works, but integrated in a bespoke way into all of the world's cloud service providers, enterprise data centers, the scale and reach necessary from our ODMs and our integrators, integrated supply chain, is vast and gigantic because the world is huge. And so, that part, we don't want to do and we're not good at doing. And -- but we know how to design the AI infrastructure, provided the way that customers would like it and lets the

ecosystem integrate it.

Well, yes. So, anyways, that's the reason why.

Operator

And your final question comes from the line of Aaron Rakers with Wells Fargo. Your line is open.

Aaron Rakers -- Analyst

Yes. Thanks for taking the question. I wanted to go back into the Blackwell product cycle. One of the questions that we tend to get asked is how you see the rack scale system mix dynamic as you think about leveraging NVLink, you think about GB NVL72 and how that go-to-market dynamic looks as far as the Blackwell product cycle.

I guess to put it simply, how do you see that mix of rack scale systems as we start to think about the Blackwell cycle playing out?

Jensen Huang -- President and Chief Executive Officer

Yeah. Aaron, thanks. The Blackwell rack system, it's designed and architected as a rack but it's sold in disaggregated system components. We don't sell the whole rack.

And the reason for that is because everybody's rack's a little different surprisingly. Some of them are OCP standards, some of them are not. Some of them are enterprise. And the power limits for everybody could be a little different.

Choice of CDUs, the choice of power bus bars, the configuration and integration into people's data centers, all different. And so, the way we designed it, we architected the whole rack. The software is going to work perfectly across the whole rack. And then we provide the system components.

Like for example, the CPU and GPU compute board is then integrated into an MGX. It's a modular system architecture. MGX is completely ingenious. And we have MGX ODMs and integrators and OEMs all over the plant.

And so, just about any configuration you would like, where you would like that 3,000-pound rack to be delivered, it's got to be close to. It has to be integrated and assembled close to the data center because it's fairly heavy. And so everything from the supply chain from the moment that we ship the GPU, CPUs, the switches, the NICs, from that point forward, the integration is done quite close to the location of the CSPs and the locations of the data centers. And so, you can imagine how many data centers in the world there are and how many logistics hubs we've scaled out to with our ODM partners.

And so, I think because we show it as one rack and because it's always rendered that way and shown that way, we might have left the impression that we're doing the integration. Our customers hate that we do integration. The supply chain hates us doing integration. They want to do the integration.

That's their value-add. There's a final design-in, if you will. It's not quite as simple as shimmy into a data center but the design fit-in is really complicated. And so, the design fit-in, the installation, the bring-up, the repair and replace, that entire cycle is done all over the world.

And we have a sprawling network of ODM and OEM partners that does this incredibly well. So, integration is not the reason why we're doing racks. It's the anti-reason of doing it. The way we don't want to be an integrator, we want to be a technology provider.

Operator

And I will now turn the call back over to Jensen Huang for closing remarks.

Jensen Huang -- President and Chief Executive Officer

Thank you. Let me make a couple more -- make a couple of comments that I made earlier again. The data center worldwide are in full steam to modernize the entire computing stack with accelerated computing and generative AI. Hopper demand remains strong and the anticipation for Blackwell is incredible.

Let me highlight the top five things, the top five things of our company. Accelerated computing has reached the tipping point. CPU scaling slows. Developers must accelerate everything possible.

Accelerated computing starts with CUDA-X libraries. New libraries open new markets for NVIDIA. We released many new libraries, including CUDA-X Accelerated Polars, Pandas, and Spark, the leading data science and data processing libraries, CUVI-S for vector databases. This is incredibly hot right now.

Ariel and for 5G wireless base station, a whole suite of a whole world of data centers that we can go into now. Parabricks for gene sequencing and AlphaFold2 for protein structure prediction is now CUDA accelerated. We are at the beginning of our journey to modernize \$1 trillion worth of data centers from general-purpose computing to accelerated computing. That's number one.

Number two, Blackwell is a step-function leap over Hopper. Blackwell is an AI infrastructure platform, not just the GPU. Also happens to be the name of our GPU but it's an AI infrastructure platform. As we reveal more of Blackwell and sample systems to our partners and customers, the extent of Blackwell's lead becomes clear.

The Blackwell vision took nearly five years and seven one-of-a-kind chips to realize, the Gray CPU, the Blackwell dual GPU, and a colos package, ConnectX DPU for East-West traffic, BlueField DPU for North-South and storage traffic, NVLink switch for all-to-all GPU communications, and Quantum and Spectrum-X for both InfiniBand and Ethernet can support the massive traffic of AI. Blackwell AI factories are building-size computers. NVIDIA designed and optimized the Blackwell platform, full stack end to end, from chips, systems, networking, even structured cables, power and cooling, and mounds of software to make it fast for customers to build AI factories. These are very capital-intensive infrastructures.

Customers want to deploy it as soon as they get their hands on the equipment and deliver the best performance and TCO. Blackwell provides three to five times more AI throughput in a power-limited data center than Hopper. The third is NVLink. This is a very big deal with its all-to-all GPU switch is game-changing.

The Blackwell system lets us connect 144 GPUs in 72 GB200 packages into one NVLink domain, with an aggregate NVLink bandwidth of 259 terabytes per second in one rack. Just to put that in perspective, that's about 10x higher than Hopper. 259 terabytes per second kind of makes sense because you need to boost the training of multitricillion-parameter models on trillions of tokens. And so, that natural amount of data needs to be moved around from GPU to GPU.

For inference, NVLink is vital for low-latency, high-throughput large language model token generation. We now have three networking platforms, NVLink for GPU scale-up, Quantum InfiniBand for supercomputing and dedicated AI factories, and Spectrum-X for AI on Ethernet. NVIDIA's networking footprint is much bigger than before. Generative AI momentum is accelerating.

Generative AI frontier model makers are racing to scale to the next AI plateau to increase model safety and IQ. We're also scaling to understand more modalities from text, images, and video to 3D physics, chemistry, and biology. Chatbots, coding AIs, and image generators are growing fast but it's just the tip of the iceberg. Internet services are deploying generative AI for large-scale recommenders, ad targeting, and search systems.

AI start-ups are consuming tens of billions of dollars yearly of CSP's cloud capacity, and countries are recognizing the importance of AI and investing in sovereign AI infrastructure. And NVIDIA AI, NVIDIA Omniverse is opening up the next era of AI, general robotics. And now the enterprise AI wave has started, and we're poised to help companies transform their businesses. The NVIDIA AI Enterprise platform consists of Nemo, NIMs, NIM Agent Blueprints, and AI Foundry that our ecosystem partners, the world-leading IT companies used to help companies customize AI models and build bespoke AI applications.

Enterprises can then deploy on NVIDIA AI Enterprise run time, and at \$4,500 per GPU per year, NVIDIA AI Enterprise is an exceptional value for deploying AI anywhere. And for NVIDIA software, TAM can be significant as the CUDA-compatible GPU installed base grows from millions to tens of millions. And as Colette mentioned, NVIDIA software will exit the year at a \$2 billion run rate. Thank you all for joining us today.

Operator

[Operator signoff]

Duration: 0 minutes

Call participants:

Stewart Stecker -- Senior Director, Investor Relations

Colette M. Kress -- Chief Financial Officer, Executive Vice President

Vivek Arya -- Analyst

Jensen Huang -- President and Chief Executive Officer

Toshiya Hari -- Analyst

Colette Kress -- Chief Financial Officer, Executive Vice President

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